

SIDDHARTH RAVIKUMAR

Senior AI Solutions Architect

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PROFESSIONAL SUMMARY

AI Solutions Architect with 14+ years building AI systems across Agentic AI, NLP, Speech Intelligence, Computer Vision, and Quantitative Intelligence. I've taken LLM-powered platforms from early prototypes through to production — dealing with the parts that don't show up in demos: failure recovery, latency budgets, cost control, observability, and keeping things reliable once real users depend on them. Most recently focused on multi-agent orchestration, enterprise RAG, and speech AI for defense applications.

CORE SKILLS

Programming Languages: Python, Java, SQL

AI/ML Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Whisper, NVIDIA NeMo

AI Specialization: Generative AI, Agentic AI, RAG Systems, Multimodal AI, NLP, Automatic Speech Recognition, Quantitative AI

Agentic AI: Multi-Agent Orchestration, LangGraph, LangChain, Tool-Calling Agents, Fallback & Self-Correction Loops, Memory-Augmented Agents, Agent Tracing & Observability

RAG & Knowledge Systems: Agentic RAG, Multimodal RAG, Hybrid Retrieval, Vector Databases (Milvus, Qdrant, Chroma), Reranking, Multi-Hop Retrieval, Knowledge Graph RAG

LLM Engineering: LLM Fine-Tuning, PEFT, LoRA/QLoRA, Open-source LLMs, Prompt Optimization, LLM Evaluation (RAGAS, DeepEval), Token Budget Management

Data Engineering & Analytics: Apache Spark, Kafka, Pandas, NumPy, Dask, DuckDB, Data Pipeline Design

MLOps & AI Deployment: MLflow, Airflow, Docker, Kubernetes, CI/CD, Model Serving, API Deployment, Structured Logging, Distributed Tracing

Architecture & Software Engineering: Distributed Systems, Microservices, Event-Driven Architecture, REST APIs, FastAPI

Cloud Platforms: AWS, GCP

PROFESSIONAL EXPERIENCE

Solutions Architect | SAAL AI *January 2025 – Present | Abu Dhabi*

Agentic Multimodal Enterprise RAG Platform

Architected and shipped an agentic document intelligence platform that goes beyond text-based RAG — handling PDFs, PowerPoints, engineering drawings, technical diagrams, and scientific documents with complex layouts, serving enterprise users daily.

- Built an Agentic RAG system using ColPali visual retrieval, OCR, Vision Language Models, and multimodal embeddings. The hard part wasn't the retrieval itself — it was figuring out why answers were wrong. Added structured logging across retrieval decisions, reranking, and LLM generation so we could trace any bad answer back to the exact retrieval failure.
- Visual document retrieval was the main bottleneck early on. Tracked grounded-answer rates per modality and iterated on the visual pipeline until we got it from roughly 62% to 84%.
- Designed multi-hop reasoning workflows with conversational memory, hybrid search, and reranking. Instrumented agent-level tracing to capture tool selection paths and fallback triggers — mostly useful for diagnosing the cases where the system went down an expensive reasoning path for a simple lookup question.

- Ran RAGAS evaluations (faithfulness, answer relevancy, context precision) on every deployment. Also tracked operational metrics: what percentage of queries resolved without human escalation, p95 retrieval latency against our 800ms target, and cost-per-query across providers to decide when to route to smaller models.

Multi-Agent Quantitative Research & Trading Intelligence Platform

Built an agentic research platform that coordinates specialized agents for market analysis, strategy evaluation, and investment intelligence. Deployed and serving real users.

- Designed the multi-agent architecture in LangGraph with agents for news, fundamentals, technical signals, and algorithmic research. Each agent logs its tool invocations, LLM calls, and token usage — not because we wanted dashboards, but because debugging a bad trade recommendation across four agents is impossible without traces.
- Added validation loops where agents cross-check their outputs and re-execute when tool calls return stale or missing data. This brought the rate of outputs needing manual review down from around 30% to under 10%.
- The agents hit multiple external APIs (Yahoo Finance, FMP, Finnhub) and any of them can rate-limit, timeout, or return garbage. Built fallback chains and TTL-based caching with market-hours invalidation to keep end-to-end latency under 3 seconds even when individual APIs are slow.
- Integrated an algorithmic trading intelligence layer (portfolio optimization, MPT, factor analysis) with feedback loops tracking whether the agent-generated signals actually held up — measuring signal consistency and Sharpe estimates against baseline over time.
- Added conversational Text-to-SQL for natural language queries over trading data and positions. Monitored SQL generation accuracy and iterated on prompt templates when we saw recurring failure patterns in the logs.

Speech Intelligence & Multilingual ASR Systems

Built production speech AI pipelines for Arabic and Persian transcription and voice intelligence. The core problem was that off-the-shelf ASR models fail badly on degraded military and operational audio — the real work was making them usable.

- Engineered a multi-tiered noise reduction pipeline for military-grade audio. Each stage addresses a different noise profile: band-pass filtering for radio and vehicle interference, spectral analysis to remove persistent frequency patterns, per-segment SNR estimation with adaptive tuning, and neural speech enhancement as a cleanup pass. Measured SNR improvement at each tier so we could skip stages on cleaner audio and avoid unnecessary latency.
- Built speech separation and clipping workflows to pull apart overlapping speakers in multi-talker military comms. Combined VAD, diarization, and segment extraction to produce clean single-speaker clips for ASR. Tracked separation quality with speaker purity scores.
- Fine-tuned Whisper and NVIDIA Conformer models on defense-domain data — military jargon, operational callsigns, tactical terms, and code words that base models consistently mangled. Ran stratified evaluations by noise level, dialect, and vocabulary type. Got over 25% WER improvement on jargon-heavy segments.
- Set up continuous evaluation with stratified test sets (clean vs. noisy, standard vs. military vocabulary, single vs. multi-talker) and automated regression checks on every model update. Also tracked how often analysts corrected transcriptions — that ended up being the most honest quality signal we had.
- The full pipeline runs as preprocessing → noise reduction → speech separation → ASR → NLP extraction. Each stage has latency and error tracking so when transcription quality drops, operators can see whether it's a noise issue, a separation failure, or the ASR model struggling with new vocabulary.

AI Architect | Ministry of Rural Development *June 2023 – January 2025 | Riyadh, Saudi Arabia*

Urban Violation Hotspot Detection & Forecasting Platform

Designed and delivered a geospatial intelligence platform used by enforcement teams to predict where urban violations will cluster, shifting their approach from reactive to planned deployment.

- Built large-scale geospatial pipelines using GeoPandas, Shapely, and PostGIS for spatial joins, proximity analysis, and location-based intelligence on city-wide datasets. Added data quality checks and processing time tracking to make sure daily batch jobs finished within their windows.
- Developed an automated hotspot detection engine with DBSCAN/HDBSCAN on GPS-tagged enforcement data. Tuned sensitivity thresholds against historical outcomes and tracked precision/recall of predictions against actual violation rates in following weeks.

- Implemented spatial-temporal forecasting models combining time-series analysis with spatial features and trend decomposition. Ran weekly evaluations comparing predictions to actual violation density, with alerts when accuracy dropped.
- The platform became the standard planning tool for enforcement teams. Tracked adoption across districts and usage frequency to confirm it was actually being used, not just deployed.

AI-Powered Municipal Compliance & Policy Intelligence Platform

Built an enterprise knowledge platform using LLMs and RAG for regulatory guidance and compliance validation, used daily by municipal operations staff.

- Implemented a multichannel conversational AI integrated with municipal backends for policy queries, compliance checks, and regulatory assistance. Logged intents, fallbacks, and unresolved queries to identify where the system was failing and feed improvements.
- Built RAG retrieval pipelines with embeddings, topic modeling, LangChain, and vector search. Tracked retrieval precision and hallucination rates across different policy domains — some regulatory areas performed much worse than others, which helped prioritize re-indexing work.
- Automated compliance checklist generation, cutting manual review time by roughly 35% as measured over a three-month operational period with the compliance team.
- Added data freshness monitoring so the system flagged when source regulations had been updated but not yet re-indexed — preventing it from serving outdated guidance.

Senior Data Scientist | Onit *September 2022 – June 2023 | Remote*

AI-Driven Legal Intelligence & Contract Analytics Platform

Built an LLM-powered platform for contract analysis, clause extraction, and legal knowledge discovery, used by legal teams to speed up review cycles.

- Developed pipelines for clause extraction, generation, and semantic validation using LLMs and retrieval workflows. Low-confidence extractions routed to human review automatically, and we tracked the intervention rate as our primary operational metric.
- Fine-tuned domain-specific legal language models with instruction tuning and contrastive learning. Reduced manual review effort by about 40%, validated through side-by-side comparison of AI-assisted vs. manual review throughput.
- Built a legal expert recommendation engine with semantic similarity, embeddings, and metadata ranking across jurisdiction and specialization. Measured acceptance rates and iterated on scoring weights based on how lawyers actually used the recommendations.
- Set up evaluation frameworks with clause-level test stratification, adversarial validation sets, and confidence scoring. Hit roughly 95% clause validation reliability with weekly regression checks to catch degradation as new contract types came in.

Consultant Data Scientist | Verizon *November 2018 – September 2022 | Chennai, India*

- Built network anomaly detection and churn prediction models using ensemble methods and sequence models (RNNs/LSTMs) running against live telemetry. Tracked alert precision and false-positive rates weekly to keep detection thresholds tuned.
- Developed NLP pipelines using BM25 and contextual embeddings to extract insights from millions of customer conversations. Had pipeline health monitoring and data quality checks to keep extraction reliable at that volume.
- Implemented sentiment analysis, intent classification, and aspect-based sentiment models. Validated accuracy on monthly holdout sets and tracked correlation with NPS score changes to measure real business impact.

Machine Learning Engineer | UST *June 2017 – December 2018 |*

- Built edge AI analytics for real-time IoT datacenter monitoring and anomaly detection, running on-premise with sub-200ms latency constraints for alerting.
- Developed streaming data pipelines on AWS (Kinesis, Spark Streaming, DynamoDB, Elasticsearch, S3, Athena) for high-volume telemetry. Handled burst traffic with backpressure monitoring and dead-letter queues.

- Developed anomaly detection models using Bayesian inference and RNN/LSTM architectures. Tracked precision/recall against labeled incidents and set up retraining triggers when detection accuracy drifted.

PERSONAL AI PROJECTS

Agentic AI Swing Trading Intelligence Platform | 202X – Present

Multi-agent quantitative research platform for systematic swing trading insights. Live at <https://app.tradesphere.space/>

- Coordinated technical, fundamental, news, risk, and trade validation agents via LangGraph. Each agent traces its tool calls, token costs, and latency — essential for figuring out which agent is responsible when a recommendation doesn't make sense.
- Built quantitative pipelines with feedback loops covering data ingestion, strategy evaluation, backtesting, and risk analytics. Instrumented pipeline stages to track data freshness, signal success rate, and end-to-end latency.
- Containerized and deployed via Docker with health checks and graceful shutdown. Monitors response times and failure rates during market hours.

On-Device Multimodal AI Image Search App | 202X

Edge AI mobile application for visual search and image understanding.

- Built computer vision pipelines with object detection, embeddings, and visual reasoning. Profiled inference latency and memory usage across device tiers.
- Optimized with TensorFlow Lite / PyTorch Mobile. Benchmarked quantized vs. full-precision models to hit both accuracy targets and 60ms latency budgets on target hardware.

RESEARCH & PROOFS OF CONCEPT

Liquid Foundation Models vs. LLMs Evaluation

- Comparative analysis of liquid neural networks vs. transformers, focused on production-relevant metrics: tokens/sec, memory per concurrent request, and performance degradation under load in low-data and streaming scenarios.

GPU Architecture & Memory Utilization Benchmarking

- Benchmarked DGX Spark vs. A100/H100 setups on VRAM utilization, bandwidth, throughput, and cost across inference, fine-tuning, and multimodal workloads. Results informed infrastructure decisions for production model serving.

Automatic Defense Action Trigger with World Foundation Models

- Developed world foundation models for defense autonomy — multimodal training across challenging physical conditions with reliability testing under sensor degradation, targeting detection consistency within strict latency constraints.

EDUCATION

Bachelor of Engineering in Computer Science

Anna University, Chennai

2007–2011